

Section 7 — Model Configuration

Concept Notes

Pydantic models have default behaviors, but sometimes an application needs different rules. Model configuration allows you to control how the entire model behaves.

In Pydantic V2, model-level settings are commonly defined using `ConfigDict` and the `model_config` attribute.

Extra Fields

Sometimes incoming data contains fields that are not defined in the model.

Pydantic can be configured to:

- Ignore extra fields
- Allow extra fields
- Reject extra fields

Using `extra="forbid"` is useful when the input structure must be strict and unexpected fields should cause validation errors.

Frozen Models

By default, model fields can be changed after the model object is created.

A frozen model prevents field reassignment after creation.

This is useful when validated data should remain unchanged throughout the application.

Validate Assignment

Normally, changing a field after model creation may bypass validation.

With `validate_assignment=True`, Pydantic validates new values whenever model attributes are updated.

This is useful when a model must remain valid throughout its lifetime.

String Transformations

Model configuration can automatically transform string values.

For example:

- Convert strings to lowercase
- Convert strings to uppercase
- Remove leading and trailing whitespace

These options help normalize input data consistently across the entire model.

Important Points

- Model configuration controls model-wide behavior.
- Pydantic V2 commonly uses `ConfigDict`.
- Configuration is assigned through `model_config`.
- `extra="forbid"` rejects unexpected fields.
- `extra="ignore"` ignores unexpected fields.
- `extra="allow"` keeps unexpected fields.
- `frozen=True` prevents field reassignment.
- `validate_assignment=True` validates values after assignment.
- `str_strip_whitespace=True` removes surrounding whitespace.
- `str_to_lower=True` converts strings to lowercase.
- `str_to_upper=True` converts strings to uppercase.
- Model configuration applies to the entire model.

Syntax

Basic Configuration

```
Import ConfigDict
```

Create Model



Define model_config



Add configuration options



Define model fields

Extra Field Configuration

```
ConfigDict(  
    extra = "forbid"  
)
```

Possible values:

ignore

allow

forbid

Frozen Model

```
ConfigDict(  
    frozen = True  
)
```

Validate Assignment

```
ConfigDict(  
    validate_assignment = True  
)
```

String Configuration

```
ConfigDict(  
    str_strip_whitespace = True,  
    str_to_lower = True  
)
```